

## POINT ARGUELLO, CA, USA

WMO: 994210

Lat: 34.5771N

Lon: 120.6470W

Elev: 106

StdP: 14.64

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 4             | 45.8       | 47.6 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 52.7                  | 52.4 | 47.5 | 52.7 | 15.2                  | 0    | 0.773 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 10            | 6.9                    | 72.6            | N/A  | 68.5 | N/A  | 65.9 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 9.0                  | 0    |

## Dehumidification DP/MCDB and HR

| 0.4%       |            |            | 1%         |            |            | 2%         |            |            | 0.4%       |            | 1%         |            | 2%         |            | Extreme<br>Max WB |
|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|-------------------|
| DP         | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth       | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (d)                        | (e)  | (f)                | (g) | (h)                                                | (i)  | (j)        | (k)  | (l)        | (m)  | (n)        | (o)  | (p)  | (q) |
| 42.4              | 35.4 | 32.2 | DB                         | 41.9 | 81.6               | 2.6 | 4.0                                                | 40.0 | 84.5       | 38.5 | 86.8       | 37.0 | 89.0       | 35.1 | 91.9 |     |
|                   |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A  |     |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 55.7   | 55.7  | 54.1  | 54.0  | 52.9  | 53.4  | 54.9  | 56.6  | 57.3  | 58.1  | 59.3  | 57.6  | 54.8 |
|                                                                                  | DBStd       | 3.95   | 4.35  | 3.86  | 3.52  | 2.71  | 2.78  | 2.09  | 2.78  | 2.50  | 3.15  | 4.64  | 4.26  | 3.54 |
|                                                                                  | HDD50       | 20     | 4     | 5     | 3     | 3     | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 3    |
|                                                                                  | HDD65       | 3411   | 291   | 306   | 342   | 363   | 361   | 304   | 260   | 240   | 210   | 190   | 228   | 316  |
|                                                                                  | CDD50       | 2113   | 179   | 120   | 127   | 90    | 106   | 146   | 206   | 225   | 243   | 289   | 229   | 153  |
|                                                                                  | CDD65       | 28     | 2     | 1     | 1     | 0     | 1     | 0     | 1     | 0     | 2     | 14    | 6     | 0    |
|                                                                                  | CDH74       | 70     | 1     | 2     | 1     | 1     | 8     | 0     | 2     | 0     | 6     | 34    | 15    | 0    |
|                                                                                  | CDH80       | 6      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 4     | 0     | 0    |
| Wind                                                                             | WSAvg       | 17.0   | 14.1  | 15.4  | 16.8  | 18.6  | 20.1  | 19.1  | 17.9  | 19.1  | 17.8  | 15.2  | 14.7  | 14.7 |
| Precipitation                                                                    | PrecAvg     | 16.6   | 3.6   | 3.7   | 2.7   | 1.1   | 0.4   | 0.1   | 0.0   | 0.0   | 0.1   | 0.8   | 1.3   | 2.8  |
|                                                                                  | PrecMax     | 32.6   | 16.3  | 15.7  | 12.6  | 4.3   | 2.4   | 0.7   | 0.3   | 0.1   | 0.4   | 4.0   | 4.5   | 9.3  |
|                                                                                  | PrecMin     | 6.2    | 0.2   | 0.1   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.2  |
|                                                                                  | PrecStd     | 6.6    | 3.5   | 3.3   | 2.8   | 1.1   | 0.6   | 0.2   | 0.1   | 0.0   | 0.1   | 0.9   | 1.1   | 2.2  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 71.6  | 70.5  | 68.7  | 66.8  | 73.1  | 62.3  | 70.4  | 65.8  | 71.4  | 78.1  | 76.3  | 66.4 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | DB     | 67.7  | 64.7  | 63.4  | 59.9  | 60.7  | 60.1  | 63.8  | 63.7  | 66.2  | 72.6  | 69.3  | 63.2 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | DB     | 64.4  | 61.2  | 60.3  | 57.7  | 58.0  | 58.9  | 61.8  | 62.5  | 63.6  | 68.0  | 65.5  | 61.3 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | DB     | 61.4  | 58.8  | 58.2  | 56.2  | 56.5  | 57.9  | 60.4  | 61.4  | 61.9  | 65.2  | 62.5  | 59.7 |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.2   | 6.0   | 5.9   | 4.8   | 4.0   | 4.0   | 3.8   | 4.2   | 5.1   | 6.9   | 7.0   | 6.2  |
|                                                                                  |             | MCDBR  | 12.3  | 11.0  | 10.9  | 8.1   | 7.6   | 5.4   | 5.3   | 5.6   | 9.0   | 14.7  | 12.4  | 8.5  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.307  | 0.319 | 0.330 | 0.343 | 0.353 | 0.351 | 0.360 | 0.344 | 0.344 | 0.342 | 0.310 | 0.311 |      |
|                                                                                  | taud        | 2.600  | 2.575 | 2.541 | 2.490 | 2.486 | 2.485 | 2.518 | 2.531 | 2.535 | 2.543 | 2.603 | 2.599 |      |
|                                                                                  | Ebn at Noon | 289    | 296   | 300   | 299   | 295   | 294   | 291   | 295   | 291   | 283   | 285   | 280   |      |
|                                                                                  | Edh at Noon | 24     | 28    | 31    | 34    | 35    | 35    | 33    | 32    | 31    | 28    | 24    | 23    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 961    | 1232  | 1664  | 2064  | 2252  | 2280  | 2121  | 1970  | 1745  | 1373  | 1067  | 856   |      |
|                                                                                  | RadStd      | 72     | 121   | 105   | 126   | 161   | 120   | 93    | 112   | 110   | 111   | 62    | 75    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (21 neighbors) | +1.08 | N/A     | -2.38  | +1.46   | N/A   | N/A   | N/A         | -292  | +401  | +119  |

Nomenclature: See separate page